

Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.50)

Metric

$$\begin{aligned} ds^2 = & -dt \otimes dt + \left(tx^2 G_3(z)^2 + ty^2 G_4(z)^2 + \frac{1}{16} (9tx^4 + 9ty^4 + 24t^2x^2 + 16t^3 + 6(3tx^2 + 4t^2)y^2)G_1(z)^2 + \frac{1}{2} (3tx^2 + 3ty^2 + 4t^2)G_1(z)G_2(z) + tG_2(z)^2 \right. \\ & + \frac{1}{2} ((3x^2 + 3y^2 + 4t)G_1(z) + 4xG_3(z) + 4yG_4(z) + 4G_2(z))\sqrt{t}G_5(z) + \frac{1}{2} (4txG_2(z) + (3tx^3 + 3txy^2 + 4t^2x)G_1(z))G_3(z) \\ & \left. + \frac{1}{2} (4txyG_3(z) + 4tyG_2(z) + (3ty^3 + (3tx^2 + 4t^2)y)G_1(z))G_4(z) + G_5(z)^2 \right) dz \otimes dz + tdy \otimes dy + tdx \otimes dx \end{aligned}$$

Coordinates

$$(x^\mu) = (t, z, y, x)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} \right) dt + \left(-\frac{3}{4} \sqrt{\frac{1}{2}} \sqrt{tx^2} G_1(z) - \frac{3}{4} \sqrt{\frac{1}{2}} \sqrt{ty^2} G_1(z) - \sqrt{\frac{1}{2}} t^{\frac{3}{2}} G_1(z) - \sqrt{\frac{1}{2}} \sqrt{tx} G_3(z) - \sqrt{\frac{1}{2}} \sqrt{ty} G_4(z) - \sqrt{\frac{1}{2}} \sqrt{t} G_2(z) - \sqrt{\frac{1}{2}} G_5(z) \right) dz$$

$$n = \left(\sqrt{\frac{1}{2}} \right) dt + \left(\frac{3}{4} \sqrt{\frac{1}{2}} \sqrt{tx^2} G_1(z) + \frac{3}{4} \sqrt{\frac{1}{2}} \sqrt{ty^2} G_1(z) + \sqrt{\frac{1}{2}} t^{\frac{3}{2}} G_1(z) + \sqrt{\frac{1}{2}} \sqrt{tx} G_3(z) + \sqrt{\frac{1}{2}} \sqrt{ty} G_4(z) + \sqrt{\frac{1}{2}} \sqrt{t} G_2(z) + \sqrt{\frac{1}{2}} G_5(z) \right) dz$$

$$m = \left(\sqrt{\frac{1}{2}} \sqrt{t} \right) dy + \left(i \sqrt{\frac{1}{2}} \sqrt{t} \right) dx$$

$$\bar{m} = \left(\sqrt{\frac{1}{2}} \sqrt{t} \right) dy + \left(-i \sqrt{\frac{1}{2}} \sqrt{t} \right) dx$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(-\frac{2\sqrt{2}}{3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \right) dz$$

$$n = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{2\sqrt{2}}{3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \right) dz$$

$$m = \left(\frac{\sqrt{2}}{2\sqrt{t}} \right) dy + \left(\frac{i\sqrt{2}}{2\sqrt{t}} \right) dx$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2\sqrt{t}} \right) dy + \left(-\frac{i\sqrt{2}}{2\sqrt{t}} \right) dx$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} \partial_t - \frac{2\sqrt{2}}{3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \partial_z$$

$$\Delta = -\frac{1}{2} \sqrt{2} \partial_t + \frac{2\sqrt{2}}{3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \partial_z$$

$$\delta = \frac{\sqrt{2}}{2\sqrt{t}} \partial_y + \frac{i\sqrt{2}}{2\sqrt{t}} \partial_x$$

$$\bar{\delta} = \frac{\sqrt{2}}{2\sqrt{t}} \partial_y - \frac{i\sqrt{2}}{2\sqrt{t}} \partial_x$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}}{4t}$$

$$\sigma = 0$$

$$\begin{aligned} \kappa = & \frac{3i\sqrt{2}xG_1(z)}{2\left(3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)\right)} \\ & + \frac{3\sqrt{2}yG_1(z)}{2\left(3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)\right)} \\ & + \frac{i\sqrt{2}G_3(z)}{3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \\ & + \frac{\sqrt{2}G_4(z)}{3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \end{aligned}$$

$$\mu = -\frac{\sqrt{2}}{4t}$$

$$\lambda = 0$$

$$\begin{aligned} \nu = & \frac{3i\sqrt{2}xG_1(z)}{2\left(3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)\right)} \\ & - \frac{3\sqrt{2}yG_1(z)}{2\left(3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)\right)} \\ & + \frac{i\sqrt{2}G_3(z)}{3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \\ & - \frac{\sqrt{2}G_4(z)}{3\sqrt{t}x^2G_1(z) + 3\sqrt{t}y^2G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{t}xG_3(z) + 4\sqrt{t}yG_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \end{aligned}$$

$$\alpha = 0$$

$$\beta = 0$$

$$\begin{aligned} \tau = & -\frac{3i\sqrt{2}xG_1(z)}{2\left(3\sqrt{tx^2}G_1(z) + 3\sqrt{ty^2}G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{tx}G_3(z) + 4\sqrt{ty}G_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)\right)} \\ & -\frac{3\sqrt{2}yG_1(z)}{2\left(3\sqrt{tx^2}G_1(z) + 3\sqrt{ty^2}G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{tx}G_3(z) + 4\sqrt{ty}G_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)\right)} \\ & -\frac{i\sqrt{2}G_3(z)}{3\sqrt{tx^2}G_1(z) + 3\sqrt{ty^2}G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{tx}G_3(z) + 4\sqrt{ty}G_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \\ & -\frac{\sqrt{2}G_4(z)}{3\sqrt{tx^2}G_1(z) + 3\sqrt{ty^2}G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{tx}G_3(z) + 4\sqrt{ty}G_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \end{aligned}$$

$$\begin{aligned} \pi = & -\frac{3i\sqrt{2}xG_1(z)}{2\left(3\sqrt{tx^2}G_1(z) + 3\sqrt{ty^2}G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{tx}G_3(z) + 4\sqrt{ty}G_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)\right)} \\ & +\frac{3\sqrt{2}yG_1(z)}{2\left(3\sqrt{tx^2}G_1(z) + 3\sqrt{ty^2}G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{tx}G_3(z) + 4\sqrt{ty}G_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)\right)} \\ & -\frac{i\sqrt{2}G_3(z)}{3\sqrt{tx^2}G_1(z) + 3\sqrt{ty^2}G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{tx}G_3(z) + 4\sqrt{ty}G_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \\ & +\frac{\sqrt{2}G_4(z)}{3\sqrt{tx^2}G_1(z) + 3\sqrt{ty^2}G_1(z) + 4t^{\frac{3}{2}}G_1(z) + 4\sqrt{tx}G_3(z) + 4\sqrt{ty}G_4(z) + 4\sqrt{t}G_2(z) + 4G_5(z)} \end{aligned}$$

$$\begin{aligned} \epsilon = & -\frac{3x^2G_1(z)}{4\left(3\sqrt{2tx^2}G_1(z) + 3\sqrt{2ty^2}G_1(z) + 4\sqrt{2t^2}G_1(z) + 4\sqrt{2tx}G_3(z) + 4\sqrt{2ty}G_4(z) + 4\sqrt{2t}G_2(z) + 4\sqrt{2}\sqrt{t}G_5(z)\right)} \\ & -\frac{3y^2G_1(z)}{4\left(3\sqrt{2tx^2}G_1(z) + 3\sqrt{2ty^2}G_1(z) + 4\sqrt{2t^2}G_1(z) + 4\sqrt{2tx}G_3(z) + 4\sqrt{2ty}G_4(z) + 4\sqrt{2t}G_2(z) + 4\sqrt{2}\sqrt{t}G_5(z)\right)} \\ & -\frac{3tG_1(z)}{3\sqrt{2tx^2}G_1(z) + 3\sqrt{2ty^2}G_1(z) + 4\sqrt{2t^2}G_1(z) + 4\sqrt{2tx}G_3(z) + 4\sqrt{2ty}G_4(z) + 4\sqrt{2t}G_2(z) + 4\sqrt{2}\sqrt{t}G_5(z)} \\ & -\frac{xG_3(z)}{3\sqrt{2tx^2}G_1(z) + 3\sqrt{2ty^2}G_1(z) + 4\sqrt{2t^2}G_1(z) + 4\sqrt{2tx}G_3(z) + 4\sqrt{2ty}G_4(z) + 4\sqrt{2t}G_2(z) + 4\sqrt{2}\sqrt{t}G_5(z)} \\ & -\frac{yG_4(z)}{3\sqrt{2tx^2}G_1(z) + 3\sqrt{2ty^2}G_1(z) + 4\sqrt{2t^2}G_1(z) + 4\sqrt{2tx}G_3(z) + 4\sqrt{2ty}G_4(z) + 4\sqrt{2t}G_2(z) + 4\sqrt{2}\sqrt{t}G_5(z)} \\ & -\frac{G_2(z)}{3\sqrt{2tx^2}G_1(z) + 3\sqrt{2ty^2}G_1(z) + 4\sqrt{2t^2}G_1(z) + 4\sqrt{2tx}G_3(z) + 4\sqrt{2ty}G_4(z) + 4\sqrt{2t}G_2(z) + 4\sqrt{2}\sqrt{t}G_5(z)} \end{aligned}$$

$$\begin{aligned}
\gamma = & \frac{3x^2G_1(z)}{4(3\sqrt{2}tx^2G_1(z) + 3\sqrt{2}ty^2G_1(z) + 4\sqrt{2}t^2G_1(z) + 4\sqrt{2}txG_3(z) + 4\sqrt{2}tyG_4(z) + 4\sqrt{2}tG_2(z) + 4\sqrt{2}\sqrt{t}G_5(z))} \\
& + \frac{3y^2G_1(z)}{4(3\sqrt{2}tx^2G_1(z) + 3\sqrt{2}ty^2G_1(z) + 4\sqrt{2}t^2G_1(z) + 4\sqrt{2}txG_3(z) + 4\sqrt{2}tyG_4(z) + 4\sqrt{2}tG_2(z) + 4\sqrt{2}\sqrt{t}G_5(z))} \\
& + \frac{3tG_1(z)}{3\sqrt{2}tx^2G_1(z) + 3\sqrt{2}ty^2G_1(z) + 4\sqrt{2}t^2G_1(z) + 4\sqrt{2}txG_3(z) + 4\sqrt{2}tyG_4(z) + 4\sqrt{2}tG_2(z) + 4\sqrt{2}\sqrt{t}G_5(z)} \\
& + \frac{xG_3(z)}{3\sqrt{2}tx^2G_1(z) + 3\sqrt{2}ty^2G_1(z) + 4\sqrt{2}t^2G_1(z) + 4\sqrt{2}txG_3(z) + 4\sqrt{2}tyG_4(z) + 4\sqrt{2}tG_2(z) + 4\sqrt{2}\sqrt{t}G_5(z)} \\
& + \frac{yG_4(z)}{3\sqrt{2}tx^2G_1(z) + 3\sqrt{2}ty^2G_1(z) + 4\sqrt{2}t^2G_1(z) + 4\sqrt{2}txG_3(z) + 4\sqrt{2}tyG_4(z) + 4\sqrt{2}tG_2(z) + 4\sqrt{2}\sqrt{t}G_5(z)} \\
& + \frac{G_2(z)}{3\sqrt{2}tx^2G_1(z) + 3\sqrt{2}ty^2G_1(z) + 4\sqrt{2}t^2G_1(z) + 4\sqrt{2}txG_3(z) + 4\sqrt{2}tyG_4(z) + 4\sqrt{2}tG_2(z) + 4\sqrt{2}\sqrt{t}G_5(z)}
\end{aligned}$$

Ricci Tensor Components

$$\begin{aligned}
\Phi_{00} = & \frac{3\sqrt{t}x^2G_1(z)}{4\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\
& + \frac{3\sqrt{t}y^2G_1(z)}{4\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\
& - \frac{t^{\frac{3}{2}}G_1(z)}{3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)} \\
& + \frac{\sqrt{t}xG_3(z)}{3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)} \\
& + \frac{\sqrt{t}yG_4(z)}{3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)} \\
& + \frac{\sqrt{t}G_2(z)}{3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)} \\
& + \frac{G_5(z)}{2\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)}
\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned}\Phi_{11} = & \frac{3\sqrt{t}x^2G_1(z)}{8\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\ & + \frac{3\sqrt{t}y^2G_1(z)}{8\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\ & - \frac{t^{\frac{3}{2}}G_1(z)}{2\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\ & + \frac{\sqrt{t}xG_3(z)}{2\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\ & + \frac{\sqrt{t}yG_4(z)}{2\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\ & + \frac{\sqrt{t}G_2(z)}{2\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\ & + \frac{G_5(z)}{4\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)}\end{aligned}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\begin{aligned}
\Phi_{22} = & \frac{3\sqrt{t}x^2G_1(z)}{4\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\
& + \frac{3\sqrt{t}y^2G_1(z)}{4\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\
& - \frac{t^{\frac{3}{2}}G_1(z)}{3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)} \\
& + \frac{\sqrt{t}xG_3(z)}{3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)} \\
& + \frac{\sqrt{t}yG_4(z)}{3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)} \\
& + \frac{\sqrt{t}G_2(z)}{3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)} \\
& + \frac{G_5(z)}{2\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)}
\end{aligned}$$

$$\begin{aligned}
\Lambda = & - \frac{t^{\frac{3}{2}}G_1(z)}{3\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\
& - \frac{G_5(z)}{12\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)}
\end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned}
\Psi_2 = & - \frac{4t^{\frac{3}{2}}G_1(z)}{3\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)} \\
& - \frac{G_5(z)}{3\left(3t^{\frac{5}{2}}x^2G_1(z) + 3t^{\frac{5}{2}}y^2G_1(z) + 4t^{\frac{7}{2}}G_1(z) + 4t^{\frac{5}{2}}xG_3(z) + 4t^{\frac{5}{2}}yG_4(z) + 4t^{\frac{5}{2}}G_2(z) + 4t^2G_5(z)\right)}
\end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"